Roots: Open Methods

Overview

- Roots
  - Points of a nonlinear equation where $f(x) = 0$
  - **Bracketing methods**: start with a guess for a range containing the root and reduce the width of the bracket iteratively. (reliable)
    - Bisection method
    - False position method
  - **Open Methods** (fast)
    - Fixed-point iteration
    - Newton-Raphson
    - Secant methods
  - Hybrid Methods: Brent's root finding method (Fast and reliable)
  - MATLAB: fzero function

Chapter Objectives

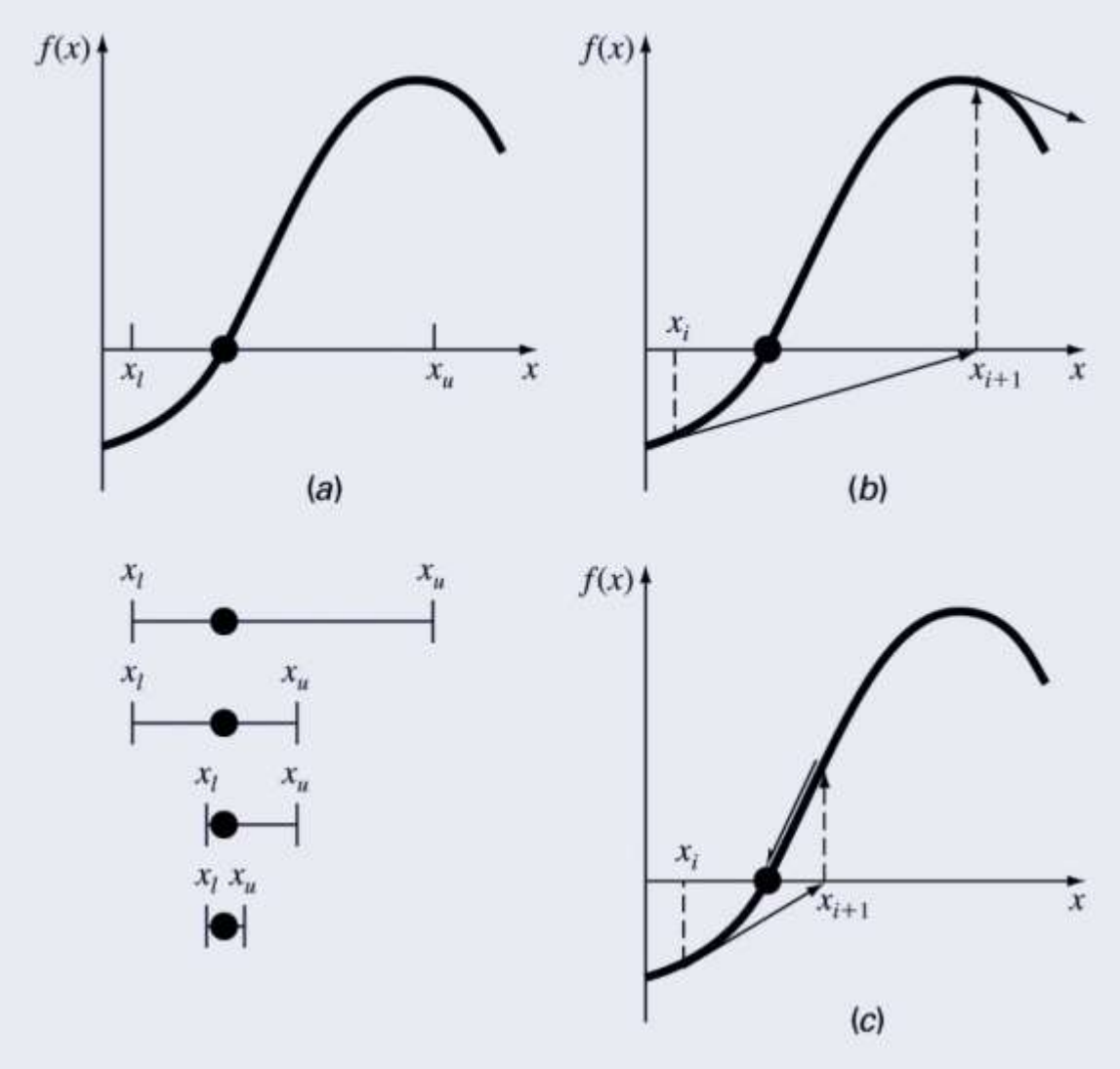
- Recognizing the difference between bracketing and open methods for root location.
- Understanding the fixed-point iteration method and how you can evaluate its convergence characteristics.
- Knowing how to solve a roots problem with the Newton-Raphson method and appreciating the concept of quadratic convergence.
- Knowing how to implement both the secant and the modified secant methods.
- Knowing how to use MATLAB's `fzero` function to estimate roots.
- Learning how to manipulate and determine the roots of polynomials with MATLAB.

Overview

- **Bracketing Methods**: always result in closer estimates of the true value of the root. (they are **convergent**)
- **Open Methods**: require a single starting point or two starting values that do not necessarily bracket the root
  - They sometimes **diverge** (move away from the root) as the computations progress.
  - When they converge, they are faster.

Graphical Comparison of Methods

a) Bracketing method
b) Diverging open method
c) Converging open method - note speed!



Fixed Point Iteration

- Also called: one-point iteration or successive substitution
- Rearrange the function $f(x)=0$ so that x is on the left-hand side of the equation: $x=g(x)$
- Use the new function g to predict a new value of x - that is, $x_{i+1}=g(x_i)$
- The approximate error is given by:

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| 100\%$$

Example

- Solve $f(x)=e^{-x}-x$
- Re-write as $x=g(x)$ by isolating x (example: $x=e^{-x}$)
- Start with an initial guess (here, 0)

i	x_i	$\|\varepsilon_a\|$ %	$\|\varepsilon_t\|$ %	$\|\varepsilon_t\|_i/\|\varepsilon_t\|_{i-1}$
0	0.0000		100.000	
1	1.0000	100.000	76.322	0.763
2	0.3679	171.828	35.135	0.460
3	0.6922	46.854	22.050	0.628
4	0.5005	38.309	11.755	0.533

- Continue until some tolerance is reached
- True value is: 0.56714329.

Example

- Solve $f(x)=e^{-x}-x$

i	x_i	$\lvert\varepsilon_a\rvert$, %	$\lvert\varepsilon_t\rvert$, %	$\lvert\varepsilon_t\rvert_i/\lvert\varepsilon_t\rvert_{i-1}$
0	0.0000		100.000	
1	1.0000	100.000	76.322	0.763
2	0.3679	171.828	35.135	0.460
3	0.6922	46.854	22.050	0.628
4	0.5005	38.309	11.755	0.533
5	0.6062	17.447	6.894	0.586
6	0.5454	11.157	3.835	0.556
7	0.5796	5.903	2.199	0.573
8	0.5601	3.481	1.239	0.564
9	0.5711	1.931	0.705	0.569
10	0.5649	1.109	0.399	0.566

Alternative Approach

- Seperate the equation into two component parts

$$f_1(x) = f_2(x)$$

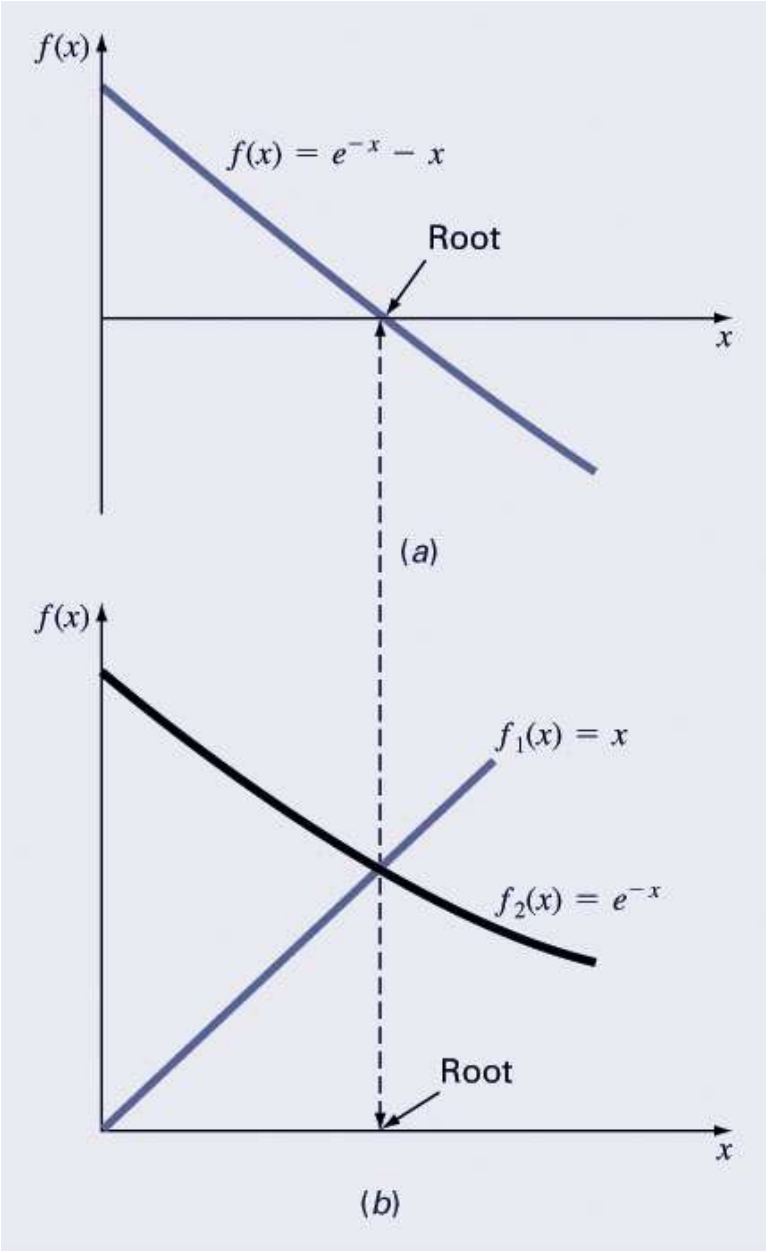
Then the two equations

$$y_1 = f_1(x)$$

and

$$y_2 = f_2(x)$$

- Can be plotted seperately. The x values corresponding to the intersections of these functions represent the roots f(x) = 0.



Convergence

- Convergence of the simple fixed-point iteration method requires that the derivative of g(x) near the root has a magnitude less than 1.
  a) Convergent, $0 \le g' < 1$
  b) Convergent, $-1 < g' \le 0$
  c) Divergent, $g' > 1$
  d) Divergent, $g' < -1$

